SAMPA STANLEY

Munyumbwe, Gwembe District, Zambia | Zambian national
stanleysampa999@gmail.com | +260 975 780 225 | linkedin.com/in/stanley-sampa-msc-129310210
Portfolio: stanleysampa999.github.io/Stanley-sampa

Professional Summary

GIS and remote sensing analyst and field ecologist based in Gwembe District, in Zambia's Middle Zambezi valley. Builds land cover, fire, river network, habitat and population maps in Google Earth Engine, ArcGIS Pro, QGIS and R from Sentinel-1, Sentinel-2, Landsat, VIIRS, MODIS, WorldPop and HydroSHEDS data. Field experience in georeferenced Miombo forest inventory, camera trapping, vegetation surveys and ODK, Kobo and SMART data collection, and has trained over 100 students and 17 professionals in GIS and SMART. MSc candidate in Tropical Ecology and Biodiversity Management at the Copperbelt University, with coursework in freshwater resources management and Kafue Flats wetland conservation.

Technical Skills

- **Remote sensing:** Google Earth Engine; Sentinel-1 SAR, Sentinel-2, Landsat, VIIRS, MODIS and Dynamic World; NDVI, burn severity, land cover classification and change detection, Random Forest
- **SMART Tool:** data model construction, database configuration, patrol design and field data collection, data validation, queries and reports
- **GIS and spatial analysis:** ArcGIS Pro, QGIS, ArcGIS StoryMaps and Dashboards; HydroSHEDS river networks, least cost path, IDW interpolation, hexagonal aggregation
- **Field data and ecology:** GPS, ODK/Kobo; forest inventory (species, DBH, tree health), vegetation surveys, camera trapping, biodiversity monitoring
- **Data analysis and reporting:** R/RStudio (species richness, abundance and trends; animal movement modelling), Microsoft Excel, thematic maps and technical reports

Work Experience

Independent GIS and Remote Sensing Consultant
Freelance **2023 to Present**

- Deliver GIS and remote sensing analysis, data management and study area maps for academic research projects.
- Mapped Zambia's 2025 population distribution in ArcGIS Pro from WorldPop 100 m data, showing settlement concentrated along the line of rail, river valleys and road corridors.
- Mapped Zambia's river network from HydroSHEDS data in QGIS.
- Mapped land use and land cover change in Gwembe District from 2017 to 2026 using Sentinel-2, Esri Land Cover and Dynamic World data, quantifying shifts in built-up area, agriculture and tree cover.
- Conducted a least cost path analysis for Nsumbu National Park using remote sensing land cover data, showing that the optimal movement route lengthened from 15 km to 71.6 km between 2013 and 2026 as land cover changed.
- Produced national maps of tree cover change and fire-related forest loss in Zambia from GLAD/UMD and FAO EarthMap (MODIS) data.

Nutri-Aid Trust, Gwembe District
District Facilitator, NAT-ZGF CSSF Project (Full time) **04/2026 to Present**

- Strengthened the governance of two farmer field schools and a Centre of Excellence in the Zambezi valley by leading executive committee renewal, constitution and by-law reviews and annual work planning.
- Train community members in governance, resource mobilisation, safeguarding, child protection and PSEA, promoting the inclusion of persons with disabilities.
- Deliver weekly, quarterly and annual reports, coordinating with community leaders, local authorities and project stakeholders.

Community Climate Solutions Zambia, Kasanka and Chitambo
Tree Inventory Intern **02/2026 to 04/2026**

- Completed tree inventory surveys on 26 georeferenced plots in open and closed Miombo woodland, recording species, DBH, tree health and ecological indicators.
- Captured and quality-checked field data with GPS, ODK and a tree inventory application, and contributed to monitoring reports and recommendations.

Zambia Agriculture Research Institute (ZARI), Kapiri Mposhi
PlantVillage Field Technician and Research and Extension Officer **04/2024 to 02/2025**

- Collected and analysed soil, rainfall and groundwater data with ODK, GIS and Ag-Observatory systems, and produced thematic maps for agricultural planning.
- Supported smallholder farmers across all 54 agricultural camps in Kapiri Mposhi District with AI-based (PlantVillage) pest and disease diagnosis and climate-smart agriculture training.

Earlier experience: Nursery and Field Assistant Supervisor, Northern Coffee Cooperation Limited, Kasama (03/2023 to 11/2023); Agronomy Industrial Attachment, ZARI Choma (10/2021 to 12/2021).

Part-Time and Volunteer Roles

Tuskline Wildlife Research and Training Institute
Training Manager (Hybrid) **05/2025 to Present**

- Train personnel in SMART Tool administration, field data collection, validation, analysis and reporting.
- Trained over 100 students and 17 professionals in SMART and in GIS analysis with QGIS and ArcGIS Pro.
- Lead camera trap and vegetation monitoring, and analyse biodiversity data in R for species richness, abundance and trends.
- Carry out habitat mapping, wildlife distribution mapping and landscape-scale vegetation analysis in Google Earth Engine.

Know Your Wildlife and the Environment Zambia (KWEZI)
Administrative Director (Hybrid) **11/2022 to Present**

- Lead administration and strategy for conservation, environmental education and community awareness work, including proposal development and stakeholder engagement.

Education

Master of Science in Tropical Ecology and Biodiversity Management
The Copperbelt University, Kitwe **06/2025 to Present**

- Modules (completed and current): Freshwater Resources Management (GTE 605), Remote Sensing and GIS (GTE 604), Climate Change and Biodiversity (GTE 602), Restoration Ecology (GTE 601), Biodiversity and Poverty Alleviation (GTE 700), Ecological Sampling Techniques (GTE 503), Quantitative Biology (GTE 501).
- Coursework: climate change adaptation and community-based conservation of the Kafue lechwe and the Kafue Flats wetland.
- Proposed research: canopy structure, fire and woodland degradation in the Gwembe Valley using GEDI LiDAR, Sentinel-1, Sentinel-2, VIIRS and field plots.

Bachelor of Science in Agroforestry (Credit)
The Copperbelt University **08/2023**

- Verified by the Zambia Qualifications Authority as a Bachelor's Degree at NQF Level 7.
- Dissertation: rainfall variability, vegetation cover and crop yield in Kasama District using remote sensing and GIS.

Professional Certifications

- Advanced Animal Movement Modelling Training Workshop (R), NITheCS, 09/2026
- Transforming Maps into Stories and Data into Decisions, Modules 1 and 2 (ArcGIS StoryMaps, Dashboards and hexagonal spatial aggregation), RCMRD/RCoE-ESA, 08/2026
- SMART for Data Analysts and Database Administrators (40 hours: data model, database configuration, patrol data and reporting), UCI/ELAP/SMART Alliance, 08/2025
- GIS Data for Agriculture and Forestry in the Open Data Era; GIS and its Application in Agriculture, DOST-PCAARRD, 09/2025
- Introduction to Forest Management Planning, FAO eLearning Academy, 06/2026
- Excel Fundamentals for Data Analysis, Macquarie University, 06/2026
- Managing Projects with Microsoft 365, MTN Skills Academy, 08/2026
- Monitoring and Evaluation, Tuskline Wildlife Research and Training Institute, 06/2024
- Fundamentals on REDD+, UNITAR/UN-REDD Programme, 12/2022
- Gender Equality, Women's Empowerment and Leadership in National Biodiversity Planning and Reporting, NBSAP Forum/UNDP, 12/2022

Leadership and Languages

- **Leadership:** Aspire Institute Leadership Program, Alumni Cohort 5 (2025); YALI Regional Leadership Centre, Business and Entrepreneurship, Alumni Cohort 20 (2024)
- **Languages:** English (proficient), Bemba (proficient), Nyanja (good), Tonga (fair)

References

Available on request.